HINDUSTHAN COLLEGE OF ARTS & SCIENCE (AUTONOMOUS)

An Autonomous Institution - Affiliated to Bharathiar University (ISO 9001-2001 Certified Institution)

Behind Nava India, Coimbatore – 641028.

DEPARTMENT OF COMPUTER APPLICATIONS (PG)

MASTER OF COMPUTER APPLICATIONS

PRACTICAL RECORD

25MCP13P – PRACTICAL:PYTHON PROGRAMMING

NAME : _____

REGISTER NO : _____

CLASS : _____

SEMESTER : _____

YEAR : _____

HINDUSTHAN COLLEGE OF ARTS & SCIENCE

(Autonomous)

An Autonomous Institution – Affiliated to Bharathiar University

(ISO 9001 – 2001 Certificate Instituation)

Behind Nava India, Coimbatore – 641028.

DEPARTMENT OF COMPUTER APPLICATIONS (PG)

CERTIFICATE

Certificate that this is a bonafide record of **PYTHON PROGRAMMING (25MCP13P)** done by _____Register No: _____during the academic year of 2025-2026.

STAFF-IN CHARGE **HEAD OF THE DEPARTMENT**

Submitted for the Bharathiar University practical Examination held on _____ at Hindusthan College of Arts & Science, Coimbatore - 28.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Date:

Place: Coimbatore

CONTENTS

S.NO	DATE	NAME OF THE PROGRAM	PAGE NO	SIGN
01		COMPUTE THE GCD OF TWO NUMBERS		
02		PERFORM TOWERS OF HONAI USING RECURSION		
03		FIND A MAXIMUM AND MINIMUM VALUE IN BINARY SEARCH TREE		
04		PERFORM SIMPLE MATRIX OPERATION		
05		PERFORM LINEAR AND BINARY SEARCH		
06		CALCULATE SUM OF ROWS,COLUMNS AND DIAGONALS USING 2D ARRAY CONCEPT		
07		PERFORM INSERTION OPERATION USING SINGLY LINKED LIST		
08		PROGRAM THAT TAKES COMMAND LINE ARGUMENT		
09		PERFORM DELETION OPERATION USING SINGLY LINKED LIST		

PROGRAM NO: 01 DATE:	COMPUTE THE GCD OF TWO NUMBERS	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
def gcd(a, b):
while b != 0:
a, b = b, a % b
return a
x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
print("The GCD of", x, "and", y, "is", gcd(x, y))
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================ RESTART: C:/Users/HP/OneDrive/Desktop/prog1.py ================
Enter first number: 12
Enter second number: 14
The GCD of 12 and 14 is 2
>>>
```

RESULT:

PROGRAM NO: 02 DATE:	PERFORM TOWERS OF HONAI USING RECURSION	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
def hanoi(n, source, helper, destination):
    if n == 1:
        print("Move disk 1 from", source, "to", destination)
        return
    hanoi(n - 1, source, destination, helper)
    print("Move disk", n, "from", source, "to", destination)
    hanoi(n - 1, helper, source, destination)
n = int(input("Enter number of disks: "))
hanoi(n, 'A', 'B', 'C')
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/pro2.py
Enter number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
>>>
```

RESULT:

PROGRAM NO: 03 DATE:	FIND A MAXIMUM AND MINIMUM VALUE IN BINARY SEARCH TREE	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
class Node:
    def __init__(self, value):
        self.value = value
        self.left = None
        self.right = None
root = Node(10)
root.left = Node(5)
root.right = Node(20)
root.left.left = Node(2)
root.right.right = Node(30)
temp = root
while temp.left:
    temp = temp.left
print("Minimum value:", temp.value)
temp = root
while temp.right:
    temp = temp.right
print("Maximum value:", temp.value)
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/pro3.py
Minimum value: 2
Maximum value: 30
>>>
```

RESULT:

PROGRAM NO: 04 DATE:	PERFORM SIMPLE MATRIX OPERATION	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
def input_matrix(r, c, name):
    matrix = []
    print(f"\nEnter elements of Matrix {name}:")
    for i in range(r):
        row = []
        for j in range(c):
            row.append(int(input(f"Element [{i+1}][{j+1}]: ")))
        matrix.append(row)
    return matrix
r1 = int(input("Enter number of rows for matrix A: "))
c1 = int(input("Enter number of columns for matrix A: "))
r2 = int(input("Enter number of rows for matrix B: "))
c2 = int(input("Enter number of columns for matrix B: "))
A = input_matrix(r1, c1, "A")
B = input_matrix(r2, c2, "B")
while True:
    print("\n1. Matrix Addition")
    print("2. Matrix Multiplication")
    choice = int(input("Enter your choice: "))
    # ADDITION
    if choice == 1:
        if r1 == r2 and c1 == c2:
            result = [[A[i][j] + B[i][j] for j in range(c1)] for i in range(r1)]
            print("\nResult (Addition):")
            for row in result:
                print(row)
        else:
            print("Addition not possible (different sizes)")
```

```
# MULTIPLICATION
elif choice == 2:
    if c1 == r2:
        result = [[0 for j in range(c2)] for i in range(r1)]
        for i in range(r1):
            for j in range(c2):
                for k in range(c1):
                    result[i][j] += A[i][k] * B[k][j]
        print("\nResult (Multiplication):")
        for row in result:
            print(row)
    else:
        print("Multiplication not possible")
```

OUTPUT:

```
*IDLE Shell 3.12.3*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/pro4.py
Enter number of rows for matrix A: 2
Enter number of columns for matrix A: 2
Enter number of rows for matrix B: 2
Enter number of columns for matrix B: 2

Enter elements of Matrix A:
Element [1][1]: 3
Element [1][2]: 4
Element [2][1]: 5
Element [2][2]: 6

Enter elements of Matrix B:
Element [1][1]: 2
Element [1][2]: 4
Element [2][1]: 6
Element [2][2]: 7

1. Matrix Addition
2. Matrix Multiplication
Enter your choice: 1

Result (Addition):
[5, 8]
[11, 13]

1. Matrix Addition
2. Matrix Multiplication
Enter your choice: 2

Result (Multiplication):
[30, 40]
[46, 62]
```

RESULT:

PROGRAM NO: 05 DATE:	PERFORM LINEAR AND BINARY SEARCH	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
def linear_search(arr, key):
    for i in range(len(arr)):
        if arr[i] == key:
            return i
    return -1
def binary_search(arr, key):
    low, high = 0, len(arr) - 1
    while low <= high:
        m = (low + high) // 2
        if arr[m] == key:
            return m
        elif arr[m] < key:
            low = m + 1
        else:
            high = m - 1
    return -1
n = int(input("Enter number of elements: "))
arr = [int(input()) for i in range(n)]
while True:
    print("1. Linear search\n2. Binary search")
    ch = int(input("Enter choice: "))
    key = int(input("Enter element to search: "))
    if ch == 1:
        res = linear_search(arr, key)
    elif ch == 2:
        arr.sort()
        res = binary_search(arr, key)
```

```
else:
    print("Invalid search")
    continue
if res != -1:
    print("Element found at index:", res)
else:
    print("Not Found")
```

OUTPUT:

```
*IDLE Shell 3.12.3*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/pro5.py
Enter number of elements: 5
6
3
8
9
2
1. Linear search
2. Binary search
Enter choice: 1
Enter element to search: 8
Element found at index: 2
1. Linear search
2. Binary search
Enter choice: 2
Enter element to search: 9
Element found at index: 4
1. Linear search
2. Binary search
Enter choice: |
```

RESULT:

PROGRAM NO: 06 DATE:	CALCULATE SUM OF ROWS,COLUMNS AND DIAGONALS USING 2D ARRAY CONCEPT	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
n = int(input("Enter size of matrix: "))
matrix = []
print("Enter matrix elements:")
for i in range(n):
    row = []
    for j in range(n):
        row.append(int(input(f"Element [{i}][{j}]: ")))
    matrix.append(row)
print("\nMatrix:")
for row in matrix:
    print(row)

# Sum of rows
print("\nSum of each row:")
for i in range(n):
    row_sum = 0
    for j in range(n):
        row_sum += matrix[i][j]
    print(f"Row {i + 1}: {row_sum}")

# Sum of columns
print("\nSum of each column:")
for j in range(n):
    col_sum = 0
    for i in range(n):
        col_sum += matrix[i][j]
    print(f"Column {j + 1}: {col_sum}")
```

```
# Sum of main diagonal
main_diag_sum = 0
for i in range(n):
    main_diag_sum += matrix[i][i]

# Sum of secondary diagonal
sec_diag_sum = 0
for i in range(n):
    sec_diag_sum += matrix[i][n - i - 1]

print("\nSum of main diagonal:", main_diag_sum)
print("Sum of secondary diagonal:", sec_diag_sum)
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/HP/OneDrive/Desktop/pro6.py
Enter size of matrix: 3
Enter matrix elements:
Element [0][0]: 2
Element [0][1]: 6
Element [0][2]: 8
Element [1][0]: 10
Element [1][1]: 15
Element [1][2]: 9
Element [2][0]: 4
Element [2][1]: 5
Element [2][2]: 1

Matrix:
[2, 6, 8]
[10, 15, 9]
[4, 5, 1]

Sum of each row:
Row 1: 16
Row 2: 34
Row 3: 10

Sum of each column:
Column 1: 16
Column 2: 26
Column 3: 18

Sum of main diagonal: 18
Sum of secondary diagonal: 27
>>>
```

RESULT:

PROGRAM NO: 07 DATE:	PERFORM INSERTION OPERATION USING SINGLY LINKED LIST	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None


class SinglyLinkedList:
    def __init__(self):
        self.head = None

    def insert_at_beginning(self, data):
        node = Node(data)
        node.next = self.head
        self.head = node

    def insert_at_end(self, data):
        node = Node(data)
        if not self.head:
            self.head = node
            return
        temp = self.head
        while temp.next:
            temp = temp.next
        temp.next = node

    # Insert at specific position
    def insert_at_position(self, data, pos):
        node = Node(data)
        if pos == 1:
            node.next = self.head
            self.head = node
            return

        temp = self.head
        for _ in range(pos - 2):
            temp = temp.next

        node.next = temp.next
        temp.next = node

    def display(self):
        temp = self.head
        while temp:
```

```
print(temp.data, end=" -> ")
            temp = temp.next
        print("None")

# Main Program
sll = SinglyLinkedList()

n = int(input("Enter number of nodes to insert at end: "))

for i in range(n):
    data = int(input(f"Enter data for node {i+1}: "))
    sll.insert_at_end(data)

data = int(input("Enter data to insert at beginning: "))
sll.insert_at_beginning(data)

data = int(input("Enter data to insert at a position: "))
pos = int(input("Enter position: "))
sll.insert_at_position(data, pos)

# Display the final linked list
print("\nSingly Linked List:")
sll.display()
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================ RESTART: C:/Users/HP/OneDrive/Desktop/pro7.py ================
Enter number of nodes to insert at end: 5
Enter data for node 1: 4
Enter data for node 2: 8
Enter data for node 3: 3
Enter data for node 4: 9
Enter data for node 5: 10
Enter data to insert at beginning: 2
Enter data to insert at a position: 7
Enter position: 2

Singly Linked List:
2 -> 7 -> 4 -> 8 -> 3 -> 9 -> 10 -> None
>>>
```

RESULT:

PROGRAM NO: 08 DATE:	PROGRAM THAT TAKES COMMAND LINE ARGUMENT	PAGE NO:

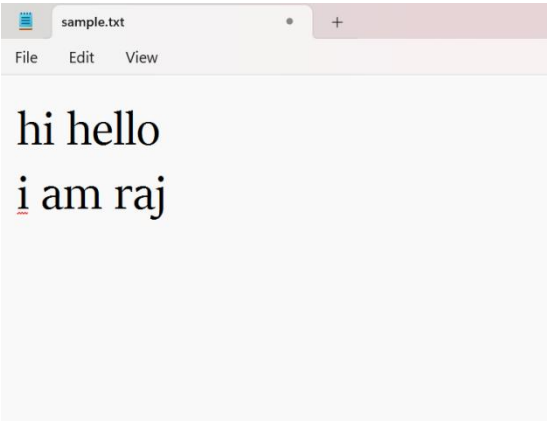
AIM:

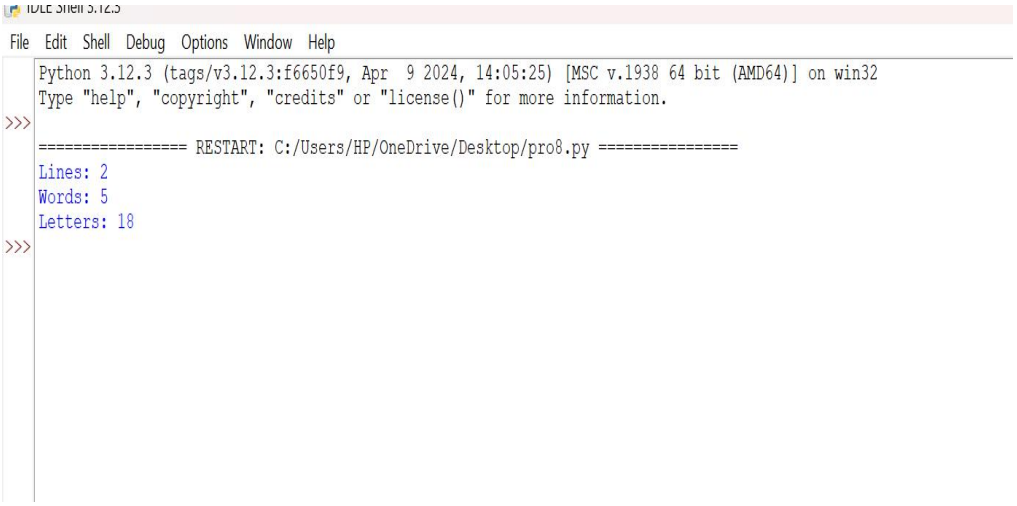
ALGORITHM:

SOURCE CODE:

```
import sys
if len(sys.argv) < 2:
    print("Usage: python file_count.py <sample.txt>")
    sys.exit(1)
fname = sys.argv[1]
lines = 0
words = 0
letters = 0
with open(fname, 'r') as f:
    for line in f:
        lines += 1
        letters += len(line)
        words += len(line.split())
print("Lines:", lines)
print("Words:", words)
print("Letters:", letters)
```

OUTPUT:





RESULT:

PROGRAM NO: 09 DATE:	PERFORM DELETION OPERATION USING SINGLY LINKED LIST	PAGE NO:

AIM:

ALGORITHM:

SOURCE CODE:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
head = None
# Insert function
def insert(d):
    global head
    new = Node(d)
    if head is None:
        head = new
        return
    t = head
    while t.next:
        t = t.next
    t.next = new
# Delete function
def delete(key):
    global head
    t = head
    prev = None
    if t and t.data == key:
        head = t.next
        return
    while t and t.data != key:
        prev = t
  t = t.next
    if t is None:
```

```
 print("Node not found")
        return
    prev.next = t.next
def display():
    t = head
    while t:
        print(t.data, end=" -> ")
        t = t.next
    print("None")
n = int(input("Enter number of nodes: "))

for i in range(n):
    insert(int(input("Enter value: ")))

print("Linked List:")
display()

delete(int(input("Enter value to delete: ")))

print("After Deletion:")
display()
```

OUTPUT:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================ RESTART: C:/Users/HP/OneDrive/Desktop/pro9.py ================
Enter number of nodes: 4
Enter value: 8
Enter value: 6
Enter value: 3
Enter value: 9
Linked List:
8 -> 6 -> 3 -> 9 -> None
Enter value to delete: 3
After Deletion:
8 -> 6 -> 9 -> None
>>>
```

RESULT: